

Random Forest Regression

Predicting Initial Hospital Days from Patient Variables

Research Question

Is it possible to predict the number of initial days spent in the hospital based on known variables in the medical dataset using the random forest method?

Analysis Goal

One goal of this data analysis is to build a random forest model capable of providing meaningful predictions of how many days patients will need to stay in the hospital to allow for the hospital to prepare accordingly.

Method Justification

The random forest model uses an average of multiple decision trees, providing a statistically robust form of prediction, hopefully ensuring high accuracy. This combination of multiple decision trees for an output is called ensemble method. Specifically, the type of ensemble method used for random forest regression is called bagging. Bagging works by randomly sampling the dataset to create multiple prediction models. These individual models are called bootstrap samples and use row sampling to attain the data to build the model. This is done using a process called bootstrapping, meaning rows are sampled with replacement. Now the model is trained, creating parallel decision trees which will run simultaneously and then aggregate their outputs into a random forest prediction.

Random forest models assume that the randomly sampled data is accurately representative of the full dataset.

Packages & Libraries

Pandas , This allows transformation of the dataset from a csv into a dataframe that can be manipulated and analyzed.

Numpy , Numpy allows for manipulation on data series, which will be necessary to calculate model metrics.

sklearn.model_selection train_test_split , This package is necessary in order to create and analyze a testing data set.

sklearn.metrics mean_squared_error , This will allow for simple calculations for MSE on the model.

sklearn.ensemble RandomForestRegressor , This contains the RFR model used for this analysis.

sklearn.metrics r2_score , This will allow for R squared calculations on the model.

sklearn.model_selection GridSearchCV , This will allow for fitting and scoring of the RFR model onto the dataset.

Seaborn , This is used to graph data distribution for outlier analysis.

sklearn.tree , Used to visualize and customize decision tree.

Matplotlib , Used to implement a heatmap for multicollinearity assessment.

Data Preprocessing Goal

The medical dataset will need to be converted to dummies and then one hot encoded before processing the model.

Variables

Numeric: Income, TotalCharge, Age, VitD_levels, Initial days

Categorical: Initial_admin, ReAdmis, Complication_risk, services, stroke, asthma, diabetes, overweight, highblood

Data Preparation Steps

Lines 13-14: First, the data is evaluated for duplicate entries using `.duplicated`, none are found.

Lines 16-36: Then, all categories in the dataset are evaluated to assess unique values, no issues are discovered.

Lines 38-39: A `.dropna` function is run over the entire dataframe to ensure any invalid values will not hold up calculations.

Lines 41-87: Outliers are evaluated for numeric columns and any outlying values are imputed with the column median.

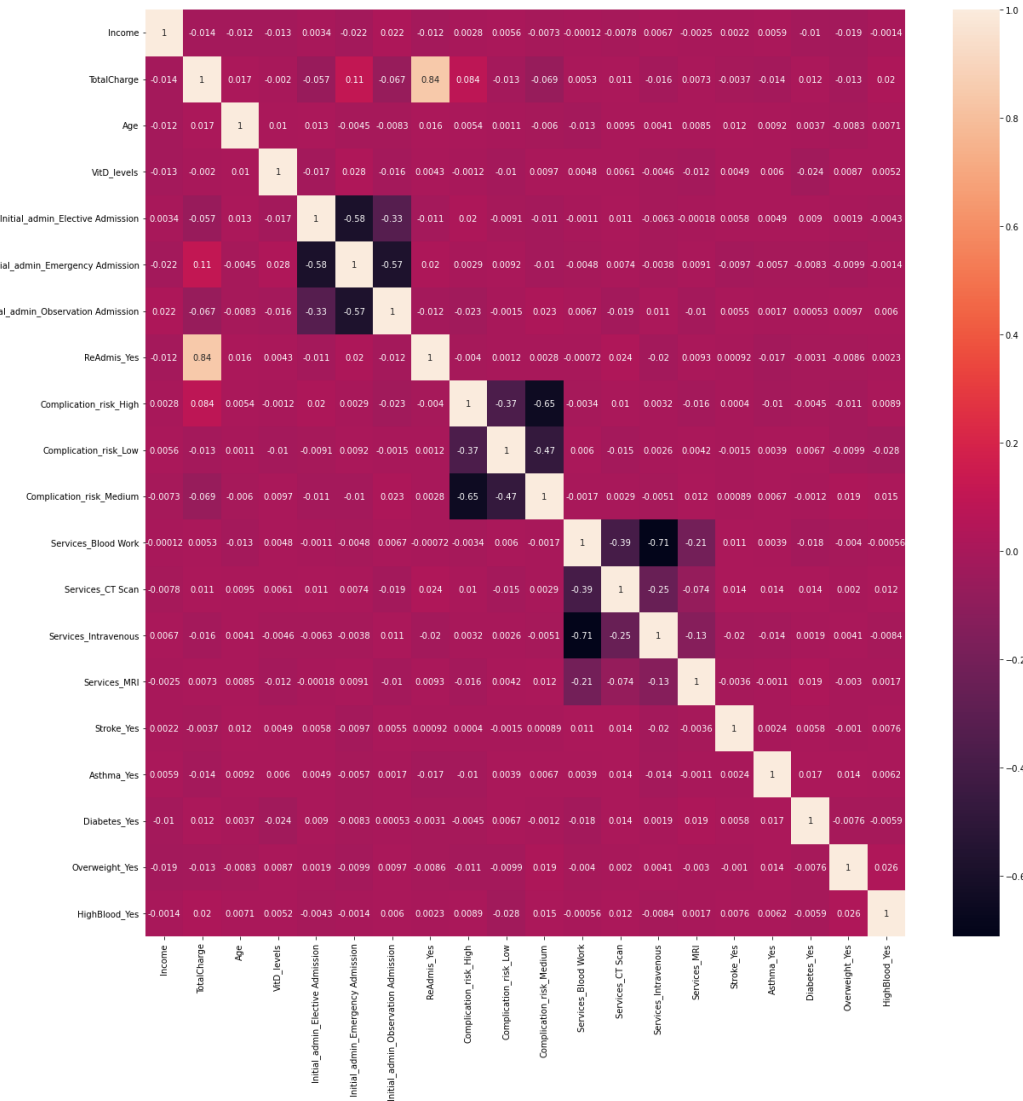
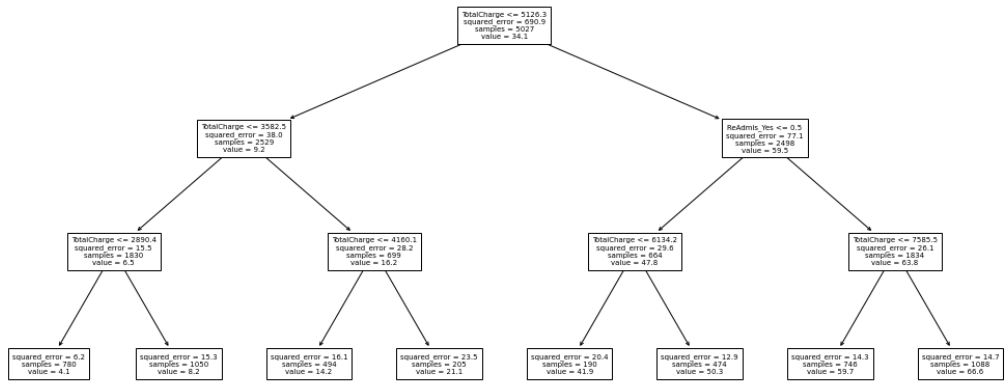
Lines 92-96: After selecting variables for analysis, the variables are transformed into dummies, and one hot encoded.

Lines 98-99: Then any binary dummy columns will have the second option removed from the dataset, as one dummy column contains all necessary information for binary variables.

Lines 101-104: After the data has been otherwise cleaned, a heatmap is created using the remaining dummy variables to assess multicollinearity. There are no variables with a strong enough correlation to suggest they should be removed.

Analysis

Random forest regression analyzes the data set by taking multiple subsets of the data and making a prediction using each subset. Then, all subset predictions are averaged to create a final prediction the model will provide.



```

Out[19]:
GridSearchCV(cv=3,
             estimator=RandomForestRegressor(max_depth=3, random_state=15),
             param_grid={'max_depth': [8, None], 'max_features': [2, 3, 4],
                          'n_estimators': [10, 50, 100]},
             verbose=3)

In [20]:
...:
...: y_pred = cv.predict(X_test)

In [21]:
...:
...: print("MSE: ", MSE(y_test,y_pred))
...: print("Root MSE: ", MSE(y_test,y_pred)**1/2)
...: print("R Square: ", r2(y_test,y_pred))
...:
...: #for potential model tuning
...: print("Best params suggestion: ", cv.best_params_)
...:
...: print("cv.best score: ",cv.best_score_)
MSE: 5.979904143861663
Root MSE: 2.9899520719308317
R Square: 0.9912199667854843
Best params suggestion: {'max_depth': None, 'max_features': 4, 'n_estimators': 100}
cv.best score: 0.9895403240882853

```

MSE & Accuracy

Mean squared error represents the averaged squared discrepancy between the predicted y values from the model and the actual y values from the dataset. Therefore, the lower the MSE, the stronger the model fits. The random forest model having a MSE of 5.97 initial days and therefore a Root MSE of 2.98 days shows that the model is, on average, outputting predicted y values that are 2.98 days away from the actual y value during testing.

This model scores an R square of 0.991 shows that 99.1% of the change in the y value, initial days, can be accounted for and predicted using the model features. This is a very strong fit and indicates the model is able to make accurate predictions within 1% of the actual y value.

Results & Implications

The MSE and R square of the random forest model are both very strong and indicate that the model is able to provide statistically relevant predictions. This model is accurate enough to be implemented for use in the hospital. Using the .best_params_ function, feature selection could be implemented on this model, further boosting its real world relevance as it could run off less data and less computing power than in this project.

Limitations

One limitation of this data analysis is that the random forest model is built using past data. The collected data is only representative of one hospital's experience over one time period, so this model may have no relevance outside of the range of data it has been trained on. There could be many unknown factors affecting how the data was distributed over the initial dataset, and those factors could easily change on the next dataset analyzed.

Course of Action

Based on the strong root MSE and R-squared scores, this random forest model should be implemented to give an initial days estimate on patient's stay in real time. The hospital now has access to a relevant prediction for how long patients will need hospital treatment, and should use that data to staff and supply the hospital accordingly. This will be most useful when the hospital experiences uncommonly high patient traffic, and can alleviate a great deal of uncertainty as to whether more staff and supplies will be needed in time to prepare accordingly.